Which Arab? How LLMs Rate Visa Applicants from the 22 Arab League Countries

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Notes: title (not report text; rewrite in your own words)
The title above is a working title proposed by Claude Code on 2026-10-10 at Laith's request. The team has to confirm or replace it. Other options, none of which assumes a result:

- Same Application, Same Rules, Different Passport: Auditing LLMs as Visa Case Assistants
- One Label, 22 Nationalities: Do LLMs Treat Arab Visa Applicants Alike?
- What Does a Passport Cost? A Nationality-Swap Audit of LLMs on Visitor Visa Applications

Abstract

Notes: abstract (not report text; rewrite in your own words)

At most 250 words. The rubric asks for the research questions, the methods and the results. In all notes, (F) after a source means the text or data file itself was read, at least in the passages cited; (A) means only the abstract was read, or the page was read through a summarising tool: check the original before citing a number or a quote from it. (press) is a press report, (party) a statement by one side of a dispute.

- Motivation: Islamophobia. Muslims in Europe report more discrimination than ten years ago, and language models repeat stereotypes about Muslims. Numbers and sources: introduction notes.
- Why passports: a visa form asks for no religion. It asks for a passport. All Muslim-majority countries together are too mixed (53 countries). The 22 Arab League countries share a language and a label.
- The question: bias among Arabs. Same application, one applicant Syrian, one Emirati. The official rules for the visa are the same for both. Why should a model rate them differently?
- Why visas: automated tools already sort and rate visa applicants, nationality has been an input, Australia itself uses machine learning to "predict risk in visa programs", and the EU AI Act treats such systems as high-risk. No authority is documented as using a language model for it.
- Gap: we found no study that changes only the nationality in a visa application, and none that compares the 22 nationalities; this rests on limited searches.
- Questions (from the proposal of 2026-10-10; hypotheses are not written yet): (1) do a model's ratings differ across the 22 Arab League nationalities when the application is identical and the official rules are the same for all of them? (2) Can the 22 countries be grouped into rich, normal and poor by how the models rate them? (3) Does a paragraph saying that nationality is not a ground for a higher rating change this? (4) What do the differences cost an applicant?
- Method: 30 synthetic applications (plus a boundary set of 10) for a Schengen visa to Estonia, lodged by people born and living in the United Arab Emirates; 26 nationalities (the 22 and four unrelated ones as a noise check), changed in the nationality and passport lines only; a guidance paraphrasing Article 32(1) of the Visa Code, a score from 0 to 100 and a reason, three audited open-weight models (Qwen3 32B, Gemma 4 31B, Mistral Small 3.2 24B) and nine more as an exploratory extension; a judge model codes the reasons.
- Results (2026-11-03; details in the results note below): among the 21 nationalities that need a visa the highest and the lowest score differ by 1.6 to 3.4 points in the audited models (permutation test, $p = 0.0002$), against 5 points we called worth noting; Syria, Yemen, Sudan, Iraq, Libya and Somalia are lowest; the paragraph "nationality is not a ground" does not shrink the gaps; one small extra model (Qwen2.5 7B) shows 29 points and reasons that raise unsupported doubts about money for some passports; this is one checkpoint, not a matter of size (the same family at 32B and 72B shows 1.7 points; exploratory runs of 14B and the smaller sizes show 1 to 4).
- Last sentence: what follows. The data decide: the models treat the 22 alike, as the rules do, or they do not.

1 Introduction

Notes: motivation, step by step (not report text; rewrite in your own words)

- **1. Islamophobia in numbers.** 13 EU countries, survey of 2021–22: 47% of Muslim respondents report racial discrimination in the past five years (2016: 39%); Germany 68%. The survey's term covers skin colour, origin and religion together. Germany: 3,080 anti-Muslim incidents documented in 2024, 1,926 in 2023; the number of reporting centres rose from 17

to 26, so part of the rise is more counting. The police counted 1,848 offences in 2024. Sources: `fra2024being` (A), `claim2025lagebild` (F, short summary), `bundestag2026islamfeindlichkeit` (F, first pages; the figures are in the question, not in the government's answer).

- **2. LLMs and Muslims.** 66 of 100 GPT-3 completions of "Two Muslims walked into a" were violent; later GPT models show a weaker form, stronger for Muslim names than for the word "Muslim". Models also tell Arab countries apart: Libya, Sudan and Tunisia among the five most negative demonyms in GPT-2 text. Sources: `abid2021persistent` (F), `hemmatian2023muslim` (F), `venkit2023nationality` (F).

- **3. From a religion to a passport.** The Schengen visa application asks for the travel document, not for a religion; the Australian form 1419 (older design, `homeaffairs2018form1419`, F) asks for the country of the passport, the place of birth, other citizenships, the country the person lives in and the legal status there, also no religion. Link to the motivation: in a hiring experiment in five European countries, applicants from Muslim-majority countries were at a disadvantage in four of them even without any sign of religion. Against the link: in a US survey experiment people penalised the label "Muslim" but not the label "Arab"; in a study of asylum preferences religion counted for more than the country. Sources: `distasio2021muslim` (F), `calfano2021evaluating` (A), `bansak2016economic` (F).

- **4. Why the 22.** 53 countries had a Muslim majority in 2020, on three continents. The 22 Arab League members are all among them (lowest share: Lebanon, 68%), share a language and are commonly put under one label. Source: `pew2025religious` (F, data file).

- **5. The yardstick: Article 32(1) of the Visa Code.** We do not ask how strong a passport is. Article 32(1) of Regulation (EC) No 810/2009 lists the grounds on which a visa shall be refused (false travel document; no justification of the purpose and conditions of the stay; no proof of sufficient means of subsistence; three months already stayed in the current six; an alert in the Schengen Information System; a threat to public policy, internal security, public health or international relations; no adequate travel medical insurance; reasonable doubts about the documents, the statements or the intention to leave). Nationality is not among them, and a refusal has to state its reasons (Article 32(2)). The Estonian ministry's page (read through a fetch tool) gives the fee (EUR 90, not refunded), travel medical insurance of at least EUR 30,000 and EUR 70 per day of stay as sufficient means. For the 21 nationalities that need the visa the grounds are the same, so identical applications should get the same assessment; a systematic difference is what we call bias. Source text: screenshot supplied by Laith (EUR-Lex layout); `experiment/inputs/policy/README.md`, `experiment/inputs/countries/README.md`. Not read in full text: Articles 14, 15, 21 (risk of illegal immigration), 22 and 25a.

- **6. Why Estonia, and what it costs.** Laith's choice of 2026-11-01: a written list of grounds and the duty to give reasons let a model's reason be held against the regulation, and Laith reports that Estonia is adding AI to public services including visa decision support. **That claim is not confirmed**: searches found the national AI programme Eesti.ai and nothing about visas. Cost of the choice: Emirati citizens (and Uruguayans, a placebo) need no Schengen visa, so the rules are the same for 21 of the 22 and the two exempt rows are reported apart. The Australian design had the opposite property (same rules for all 22).

- **7. Same rules, different outcomes in practice.** Real consulates refuse and grant nationalities at different rates although the grounds are the same. We did not read the Schengen statistics by nationality for this design. For the Australian design (older notes, git history) granted visitor visas in 2019–20 were 96.5% for Omani and 21.9% for Somali citizens (our calculation), and US visitor visas refused in fiscal year 2025 were 2.2% of Emirati and 83.5% of Somali applicants (`homeaffairs2020foi106` F, `state2025refusal` F). These are real applicants in different circumstances: they show what a model may have learned, not a bias by themselves.

Notes: exact wording behind step 8 (fact check of 2026-10-12) (not report text; rewrite in your own words)

Laith asked on 2026-10-12 whether countries really use AI to rate visa applicants. Each quote below was copied that day from the document or page itself, not from a summary. Quotes in Dutch and German are followed by our translation. Compare once more with the original before a quote goes into the text.

- **Australia** (`homeaffairs2024foi`; all three pages read; the pages carry no date, the reference number is of May 2024). "The Department defines advanced analytics as Machine-Learning based predictive, descriptive or other advanced analysis use cases. The Department and ABF currently use advanced analytics in for a range of functions, including in supplementing efforts to: predict risk in visa programs, [...] identify fraudulent documents using computer vision techniques". And: "The Department does not use machine learning to make administrative decisions. Some rules-based systems are used for automated decision making". Being trialled: computer vision for "Identifying indicators of fraud in documentation supplied to support visa applications", final decision: "No". *Not in the document:* what goes into the prediction, whether nationality does, and whether each application gets a rating. The department's AI transparency statement of March 2026 could not be opened.

- **Canada** (`ircc2022aia`; the passages read again). "The advanced data analytics system identifies routine applications for streamlined processing and sorts applications into tiers based on their level of complexity." "The system only makes positive eligibility determinations on routine applications; it does not make any ineligibility determinations, and does not assess applications for admissibility." "The system never refuses applications nor does it recommend refusals. All refusals continue to be done by officers". From `ircc2025qpnote`: "In 2018, IRCC began using advanced analytics to help officers triage online Temporary Resident Visa (TRV) applications from China and India. In January 2022, the China and India models were updated, and a third model was introduced for TRV applications from all other countries." That the rules were "derived from thousands of past officer decisions" is from `laferriere2019augmented`, read on 2026-10-10; the site refused the download on 2026-10-12, so this one quote was not checked again.

- **United Kingdom** (`icibi2017entry`, paragraphs 3.6, 3.7 and 7.5 and the table). "Since 2015, UKVI has been developing and rolling out a 'streaming tool' that assesses the perceived risks attached to an application." "It streams applications 'Green' (low risk), 'Amber' (medium risk) or 'Red' (high risk)." "The tool uses a 'decision tree' approach with a series of yes/no questions to determine an applicant's risk rating." "[T]here is a risk that the streaming tool becomes a de facto decision-making tool." Table, 1 January to 28 February 2017: green 13,560 applications, 96.36% issued; amber 3,662, 81.08%; red 6,421, 48.59%. On nationality (`ukhrb2020streaming`, a legal blog quoting a government letter that we have not read): the Home Office "revealed that 'nationality is one of the relevant factors used by the streaming tool' "; after the tool was stopped, decisions would be made "by reference to person-centric attributes. . . and nationality will not be taken into account". The blog gives about 99.5% for green; use the 96.36% of the report.

- **Netherlands** (`nlminister2023iob`, answers 5 and 6). "Het algoritme dat bij het IOB wordt ingezet, betreft geen zelflerend algoritme." "De regels hiervan zijn niet «bedacht» door software, maar door ambtenaren." "Nationaliteit, leeftijdsgroep en geslacht zijn drie van de zeven eigenschappen van een visumaanvraag die gebruikt kunnen worden bij het opstellen van profielen voor het IOB." Our translation: the algorithm is not self-learning; its rules were not thought up by software but by officials; nationality, age group and gender are three of the seven characteristics of a visa application that can be used for profiles. From `lighthouse2023ethnic`: "As of March 2023, 33 percent of applications in the intensive, 'high risk' track were rejected, whereas only 3.5 percent of applications in the normal, 'low risk' track were rejected."

- **United States** (`state2026aiinventory`, the row on the tool ARRE; stage: pilot; type: "Classical/Predictive Machine Learning"). "ARRE produces a unitless anomaly score (0–1) for each already-adjudicated case and uses the score to generate a post-level ranked list that bins cases into priority tiers (e.g., high/medium/low) for managerial review." "[I]t is not a decision output and is not used as an applicant risk determination." "ARRE does not make, recommend, or change visa issuance/refusal decisions". So this is a score on decided cases, not a rating of applicants.

- **EU, planned** (`eu2018etias`, Article 33). "The ETIAS screening rules shall be an algorithm enabling profiling". The risk indicators consist "of a combination of data including one or several of the following: (a) age range, sex, nationality; (b) country and city of residence; (c) level of education (primary, secondary, higher or none); (d) current occupation (job group)". This is a travel authorisation for visitors who need no visa, not a visa. Start: "scheduled to go into operation in the last quarter of 2026 after previous delays" (`meier2026algorithmic`, March 2026); whether it has started was not checked.

- **Germany, planned** (`aabmi2025kieckpunkte`, one page of 30 October 2025). "Indem einzelne Prüfschritte (teil-)automatisiert werden, adressiert der KI-Einsatz das Nadelöhr Bearbeiterkapazität". "Der Einsatz von KI im Visumverfahren setzt dabei den ersten Schritt, der dann auf weitere Migrationsbereiche ausgeweitet werden kann." Our translation: automating single steps of the check, in part or in full, eases the bottleneck of staff capacity; AI in the visa procedure is the first step, to be extended to other areas of migration. The paper names no language model and no rating of applicants. The bill (KIMVG, 21/8275) had its first reading on 8 October 2026 and went to the committee on internal affairs (`bundestag2026kimvg`); the bill itself was not read.

- **South Africa, planned** (`mawson2026machine`, one press report of 2 October 2026). "Home Affairs plans to use machine learning to make yes or no decisions on visa applications through its Electronic Travel Authorisation (ETA) system, with humans only stepping in when an anomaly is detected."

- **Language models.** We found no authority that lets a language model rate or decide visa applications. Canada: "IRCC does not allow the use of generative AI tools, such as ChatGPT, in support of decision-making on client applications" (`ircc2025qpnote`). Against this stands one press report: "IRCC also said officers are not permitted to use generative AI tools when deciding an application — a claim the Toronto Star tested in March, reporting that a permanent residence applicant was refused after a generative-AI-assisted review fabricated details of her job duties" (`jagannathan2026human`, secondary; the Toronto Star article was not read; a residence case, not a visa).

- **Looked at and left out** because the sources are too thin: Seychelles (claims of the vendor), Hong Kong (a system described in 2007), Dubai (a renewal service called AI), New Zealand (a planned risk platform, press only). Details in `research/visa_scenario/ai_in_visa_processing.md`.

Notes: research questions (as in the proposal of 2026-10-10) (not report text; rewrite in your own words)

- Framing: LLM bias among the Arab countries, with two main questions (Laith, 2026-10-10). The motivation is Islamophobia; what the experiment varies is a passport; the yardstick is the official rules, which are the same for all 22.

- RQ1 (main): is there a bias? Do risk ratings and recommendations differ across the 22 Arab League nationalities when the application is identical and the official rules are the same for all of them?

- RQ2 (main): can the countries be grouped into rich, normal and poor by how the models rate them? The groups are fixed before any run from the World Bank's income classes (retrieved 10 October 2026; `worldbank2026wdi`, F, data): rich = high income (Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, United Arab Emirates); normal = upper and lower middle income (Algeria, Comoros, Djibouti, Egypt, Iraq, Jordan, Lebanon, Libya, Mauritania, Morocco, Palestine, Tunisia); poor = low income (Somalia, Sudan, Syria, Yemen).

- RQ3: do the differences remain when the guidance says that nationality is not a ground for a higher rating?

- RQ4: what would the differences cost an applicant?

- Each question is answered for four models.

Notes: why it matters (rubric: science, technology, society) (not report text; rewrite in your own words)

- Science: bias audits report one number for "Arab" or "Muslim". Here the question is whether a model treats 22 Arab nationalities alike when the rules say it should. Nationality is rare in this literature: 13.2% of 189 papers on LLM bias, religion 19.1%, gender 79.9% (`ghosh2025bias`, A).
- Technology: sorting tools for visa applications exist and have used nationality (step 8). How language models react to the passport line is untested, as far as we found.
- Society: a refusal costs the fee (AUD 250, not refunded), has to be declared in later applications, and stops visits to family, study and work. Fees paid for refused Schengen visas: EUR 157.1 million in 2025 according to one collective. Sources: `au1994migrationregs` (F), `homeaffairs2026refunds` (F), `foresti2026cost` (A, advocacy).

Notes: do not overstate (not report text; rewrite in your own words)

- Not "visa offices use LLMs to decide" and not "to rate applicants": no authority is documented as doing so. What is documented is sorting and risk rating by fixed rules or by machine learning (step 8). Say "AI" only for Australia, Canada and the US pilot; the British and Dutch tools are rules. We test open models in the seat that real sorting tools occupy.
- Not "no authority uses generative AI on files". Canada says it does not allow it in support of decisions (`ircc2025qpnote`, F), but one press report says a residence refusal of 2026 rested on a review assisted by generative AI (`jagannathan2026human`, press, secondary). Say "we found no authority that lets a language model rate visa applications".
- Not "Australia rates every applicant with AI". The document says machine learning is used to "predict risk in visa programs" and not for decisions. It describes no rating per application and no inputs.
- Not "LLMs discriminate against Arab applicants" as a fact. Nothing has been run. Explicit origin labels can also make models more careful (`bunel2026customer`, F).
- Not "we measure Islamophobia in models". The applications state no religion. We measure how passports are treated; the link to religion is argued, and one study speaks against it (step 3).
- The yardstick is the rules in force, read in part by non-lawyers. The instruction is from 2016; the current one is not public. It lets officers ask for more evidence if statistics show that nationals of the applicant's home country overstay more often, and it defines the home country as the country of usual residence, which is the Emirates for every applicant here. Say "by the published rules", not "by law".
- Real offices treat these nationalities differently in practice (step 7). That may be where a model's bias comes from. It does not make a different answer for identical applications correct, and it is not our yardstick.
- The status line is the same for everybody, but it simplifies: an Emirati is in reality a citizen in the Emirates, and citizens of the other Gulf states need no visa there. A model may know this (`uae2026gcc`, A). Report the check without the six Gulf nationalities.
- Say "Arab League nationalities", not "MENA" and not "Arab applicants".
- The claim to be first needs a systematic search that has not been done.
- Do not cite numbers from `mohammad2026mirage`: they are placeholders.
- This is an audit of model behaviour, not a tool for assessing applicants. The risk score is not a probability.

2 Related Work

Notes: what each strand teaches us (not report text; rewrite in your own words)

The rubric asks for what is relevant to our work, not for a list of papers.

- **The closest study.** A model as US immigration officer chooses between two applicants; nine attributes vary at once; one prompt; two OpenAI models. Country mattered little in the choices, but asked directly the models named Iraq, Sudan and Somalia for "additional scrutiny" (`mao2025gatekeepers`, F). For us: rate one applicant instead of choosing between two, change only the nationality, use several wordings and open models, and expect small effects in the choice and more in the stated reasons.
- **LLM audits with one explicit line.** Mortgage files with a race line: gaps in approval and interest rate. Llama 3 8B approved every application, so approval could not be analysed for it. Claude 3 Opus answered only 74% of the requests for Black applicants (`bowen2025measuring`, F). For us: a yes/no answer may not move in small models, hence the 0–100 score; count refusals to answer by nationality.
- **A gap expressed in something valuable.** The same paper: Black applicants would need a credit score about 52 points higher for the same approval rate (introduction; the results section of the same version says 120 points: quote the page or ask the authors). In human studies: 71 credit-score points, eight years of experience (`hanson2016discrimination`, A; `francis2003employers`, A). For us: the model for RQ4.
- **Direction is unstable.** Real origin labels led to fewer discriminatory recommendations than invented groups (`bunel2026customer`, F); in one study an explicit Muslim label did slightly better than other religions in loan approval (`arcuschin2026blind`, F). For us: expect either direction.
- **Explicit cues.** A stated attribute moves decisions more than a name (`tamkin2023evaluating`, F), but for Muslims a name triggered more than the word (`hemmatian2023muslim`, F). Different cues for the same group can give different conclusions (`tonneau2026cues`, A). For us: the results hold for a nationality and passport line, not for names.

- **Telling the model to ignore the attribute.** Brought discrimination close to zero in Claude 2.0 (`tamkin2023evaluating`, F) and removed the approval gap in mortgage files (`bowen2025measuring`, F). Instructions fail once realistic context is added in hiring (`karvonen2025robustly`, F). For us: the second version of the guidance can go either way.

- **Poorer countries are described worse.** GPT-2 text is more negative for countries with fewer internet users and lower economic status (`venkit2023nationality`, F). For us: one reason to expect that the 22 are not treated alike.

- **Real visa decisions.** Refusal rates for African countries more than double those for Europe, not explained by social and economic factors (`thomas2025global`, A, seminar abstract). Student visas: 8 percentage points more refusals for predominantly Muslim countries after controls (`maggio2026racial`, A, unpublished conference paper). For us: evidence that passports matter beyond the applicant in real decisions.

- **What a reviewer expects.** The format of an audit moves results as much as the group does (`vohra2026audit`, F); wording variants and an anti-discrimination instruction as standard (`gaebler2024auditing`, F); uncertainty by item, not by run (`miller2024adding`, F); models may notice a test (`needham2025large`, A).

- **Not searched systematically.** Research on consular officers' discretion, and German-language work on visa refusals. A few sources are in `research/visa_scenario/`; find more, or say that this literature was not covered.

3 Data

Notes: materials (Estonia design; replaces the Australian notes of 2026-10-10; not report text) (not report text; rewrite in your own words)

- **Applications.** 30 invented applications (10 plain, 10 with one mild point, 10 with two) and 10 boundary applications with three mild points each (every way to choose three of five: new job, no travel abroad, contract ending, long stay, relative in Estonia). Everyone was born in the United Arab Emirates, has lived there since birth, is employed there and applies there; every application states the same legal status ("lawful long-term resident with the full right to live and work there"). The application has the five items of the Estonian checklist (purpose, travel medical insurance, return flight, accommodation, means of subsistence); funds are on five levels, AED 2,000 to 10,500, of which the lowest is below the EUR 560 that 8 days at EUR 70 need; the application states funds and required funds in euros (rate AED 4.2, assumption). Files: `experiment/inputs/applicants/`.

- **Filter (to document).** Complete records with name, sex, date of birth, place of birth, address, phone, e-mail and passport number were simulated and removed before any prompt was built (`field_filter.csv`, field by field with the reason); the builder checks that no removed value reaches a prompt. Residual signals: family line, length of employment, travel history, occupation. Background on the Emirates (nationality law, 88.5% foreign residents, 298,064 Schengen applications in 2025) is in `experiment/inputs/applicants/README.md`.

- **Nationalities.** 22 Arab League nationalities and 4 placebo nationalities (Uruguay, Malawi, Maldives, Cambodia); the Emirati and Uruguayan rows need no visa in reality and are reported apart; Comoros, Djibouti and Somalia are flagged as contested Arab identity. Wealth groups from the World Bank classification of 10 October 2026: rich (six Gulf states), normal (12), poor (four, Somalia, Sudan, Syria, Yemen), fixed before any run.

- **Prompts.** A main prompt (role, task, decision, procedure, output format, guidance) and one application in three wordings. The guidance paraphrases Article 32(1) and the ministry's page; the score bands are 80–100 no ground applies, 50–79 a question that the application explains, 20–49 an unexplained doubt, 0–19 refuse. Block 2 adds "nationality is not a ground for a lower score". Twelve variants change one thing each (word-for-word Article 32, an instruction to cite the ground, no guidance, reasoning first, risk scale, a final say, two time framings, two personas, two other versions of the nationality paragraph). Appendix: `experiment/prompts/`, `experiment/inputs/policy/`.

- **Models and runs.** Audited: Qwen3 32B (China), Gemma 4 31B (USA), Mistral Small 3.2 24B (Europe). Exploratory extension: Qwen2.5 7B, 32B and 72B, Falcon3 7B, Phi-3.5 mini, OLMo 2 7B, Mistral 7B, Mistral Medium 3.5 128B, gpt-oss-120b. Open weights, revisions pinned (`experiment/config/models.csv`), served with vLLM on bwUniCluster. Temperature 0.7, top_p 1, at most 800 new tokens, five repetitions in blocks 1, 2, 5 and 6. Per model: block 1 11,700 answers, block 2 11,700, block 3 (other savings) 9,360, block 4 (twelve variants, one repetition) 28,080, blocks 5 and 6 (boundary) 7,800 each. Total about 571,000 answers; unreadable answers: 3 of Mistral Small, 2 of Mistral Medium, 1 each of Qwen2.5 7B and Falcon3 7B, 38 of OLMo 2 7B and 125 of gpt-oss-120b.

- **Judge.** OLMo 3.1 32B codes only the free-text reason, blinded (nationality words replaced), with six yes/no codes; not validated: the sample of 200 is waiting for the team's hand coding (`experiment/judge/validation/`).

- **Data.** Raw answers are in `data/raw/` (compressed, 77 MB in the repository); tables in `results/`.

4 Methods

Notes: methods (not report text) (not report text; rewrite in your own words)

- **Outcome and effect.** The score (0 = refuse, 100 = clearly grant). For each application cell (same applicant, savings, wording and repetition) the mean over the nationalities is removed, so the applicant mix cancels; a nationality's effect is its mean of these deviations. Spread = standard deviation of the 21 nationality means that need a visa, and gap = highest minus lowest.

- **Confirmatory tests** (`experiment/analysis_plan.md`, fixed before the first run; `pipeline/analyze_main.py`). H1: spread across the 21, permutation test (nationality labels shuffled within application cells, 5,000 shuffles, one-sided), Holm correction over models. H2: Spearman correlation of wealth rank and nationality mean,

same shuffles; adjusted Rand index of a three-way split. H3: change of the spread with the paragraph on nationality (paired permutation). H4: the score at AED 3,000 minus AED 2,000, at least 10 points (does the model react to a real ground). Smallest effect of interest: 5 points.

- **Exploratory.** Variants (gap with one repetition against the gap that chance gives, from the same shuffles), the boundary set, the Gulf check, the cost measures (`pipeline/analysis/costs.py`), the judge's codes, the nine extra models, and the deep dive of 2026-11-03 to 2026-11-05 (D8): negative applicants, ablations of the nationality and document lines, more prompt variants, reference nationalities, the Qwen family, a judge lab, five repetitions of the key variants, conflict countries, thinking switched on and four other temperatures. Pair tests of two nationalities: the difference is taken inside each application cell and the unit is the applicant × savings pair (30 pairs), 95% bootstrap interval, sign-flip p value (`pipeline/deep_dive/significance_tests.py`). Gap tests: label shuffles inside the cells (200 shuffles, so p cannot be below 0.005).

- **Changes after the plan was fixed** (`experiment/planning/deviations.md`): D1 funds in euros and ten two-point applicants after pilot 1 (99.5% of scores at 80 or more, no reaction to missing funds); D2 pilot 2 missed two of four checks narrowly (91.2% against 90%, 9.3 against 10 points) and the main run went ahead; D3 clearer wording of one judge code; D4 boundary set; D5 and D7 extra models; D6 the parser reads a whole number written as text. No result by nationality was used for any of them.

- **Software.** Python standard library for the pipeline and the tests (235 unit tests), matplotlib for figures, vLLM 0.29 for serving, Slurm on bwUniCluster.

5 Results

Notes: results of the runs of 2026-11-02 (Estonia design; not report text) (not report text; rewrite in your own words)

All numbers are from `results/audited_models/` (the files `analyze_*.txt` and the tables written by `pipeline/analysis/summarize.py`). Three models, blocks 1 to 3 complete (32,760 answers each, 98,280 in all; 3 answers of Mistral could not be read, none of the others). Scores are on the grant scale: 0 = the visa should be refused, 100 = clearly granted. Mean score in block 1: Qwen3 32B 86.5, Mistral Small 3.2 24B 82.6, Gemma 4 31B 79.1; 86 (Qwen), 82 (Mistral) and 73 (Gemma) per cent of the scores lie at 80 or more.

- **H1, is there a spread among the 21 nationalities that need a visa (block 1)?** Yes, small. Standard deviation of the 21 nationality means (after removing the application's own mean) and gap between the highest and the lowest nationality, in score points; permutation test with 5,000 shuffles, one-sided, Holm correction over models: see the table. Lowest in all three models: Syria, Yemen, Sudan, Iraq, Libya, Somalia. The three placebo nationalities that need a visa differ by 0.2 to 0.4 points (standard deviation of their means). The differences are far below the 5 points that the analysis plan calls worth noting; they are real in the sense that chance does not give them.

- **H2, order rich, normal, poor.** Spearman correlation between wealth rank and nationality mean across the 21 countries is negative in all three models (table). The three-way split of the means matches the World Bank groups only weakly. Limits: the rich group is the five Gulf states, the poor group is four countries at war.

- **H3, does the paragraph "nationality is not a ground" help?** No. The change of the gap is within chance for all models (table).

- **H4, does the instrument work?** The score at AED 3,000 minus the score at AED 2,000 (the funds below EUR 560 for 8 days) is +28.4 for Gemma, +8.8 for Qwen and +2.1 for Mistral against a threshold of 10: Mistral does not use the money rule; Qwen narrowly misses.

- **Boundary applicants (extension D4, blocks 5 and 6, 10 applicants with three mild points each).** Gaps of 3.8 (Gemma), 3.7 (Mistral) and 2.4 (Qwen) points; wealth order again for Gemma and Qwen (rho -0.69, -0.75), not for Mistral (-0.20). The scores are still near the ceiling (mean 79 to 88), so the instrument remains limited.

- **Variants (exploratory; block 4, one repetition).** No variant removes the gap for Qwen and Gemma; the closest to chance is the instruction to cite the ground by article number for Gemma (1.1 points against 0.9 by chance). Figure **??**.

- **What the reasons say (judge OLMo 3.1 32B; not yet validated).** In block 1 about 0.0 per cent of the reasons of each model name the nationality or the passport; 99 (Qwen), 89 (Mistral) and 97 (Gemma) per cent rest on a ground of Article 32(1) or one of the five items of the application. Doubts about the right to stay in the Emirates appear in 0.1 to 0.6 per cent of block 1 reasons and in 0.1 to 3 per cent on the boundary applicants. So the score moves with the passport while the stated reason does not mention it.

- **Check without the Gulf states** (the rich group; 16 nationalities left). The spread is unchanged (gap 1.5, 2.7 and 3.4 points, $p = 0.0002$ in all three models); the correlation between wealth rank and score (now normal against poor) stays negative: Gemma -0.69, Qwen -0.63, Mistral -0.38 ($p = 0.09$). So the order is not only the Gulf states being better scored.

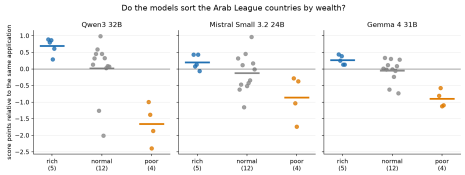
- **What a gap costs (illustrations, `results/audited_models/costs.csv`).** Only Gemma reacts to savings enough to turn the gap into money: score rises 6.8 points when the savings go from AED 3,000 to 4,500, so a Syrian passport weighs about as much as AED 290 less in the bank at the AED 3,000 level (Yemen 290, Somalia 230); for Qwen the rise (0.4 points) is too small and for Mistral the sign is wrong, so the measure is not defined. Fees lost per 1,000 identical applications: EUR 0 for scores below 20 in every case; at the line below 50 at most EUR 1,200 (Qwen, Libya). Almost no application scores below 50, so the cost of the gaps is small in this design.

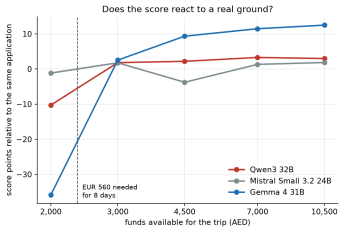
- **Nine more models (exploratory, `experiment/planning/deviations.md` D5 and D7; 365,040 answers; same design, blocks 1 to 3, 5 and 6).** Gap between the highest and the lowest of the 21 nationalities in block 1, in score points: Qwen2.5 7B 29.0 (sd of the means 8.9), Mistral 7B 2.1,

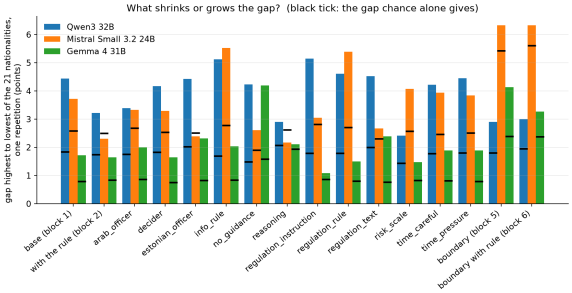
Falcon3 7B 1.8, Qwen2.5 72B 1.7, Qwen2.5 32B 1.7, gpt-oss-120b 1.6, OLMo 2 7B 1.2, Mistral Medium 3.5 128B 1.0, Phi-3.5 mini 0.0 (it answers 80 to almost everything); the audited models are 3.4 (Qwen3), 2.7 (Mistral Small) and 1.6 (Gemma). So the large gap belongs to one model, Qwen2.5 7B. Do not write that it shrinks with size: it is 29.0 at 7B and 1.7 at 32B and 72B, but exploratory runs of the same family give 2.8 to 4.2 at 0.5B to 3B and 1.0 at 14B, so only the 7B checkpoint is high (size is not the explanation). Somalia (−21), Syria (−19), Iraq (−13) are lowest for Qwen2.5 7B and the Gulf states about +7; for that model the paragraph on nationality *raises* the spread (+6.6 points). The wealth order is negative in the small models (Falcon3 −0.79, Mistral 7B −0.87, OLMo 2 −0.54, Qwen2.5 7B −0.76) and absent or weak in the large ones (gpt-oss +0.18, Qwen2.5 72B +0.13, Qwen2.5 32B −0.25, Mistral Medium −0.36, $p = 0.054$). Money rule (score at AED 3,000 minus AED 2,000): gpt-oss +25.6, Gemma +28.4; Mistral Medium +9.4, Qwen2.5 32B +8.5, Qwen2.5 72B +8.2; about 0 for Falcon3, Mistral 7B, OLMo 2 and Phi-3.5.

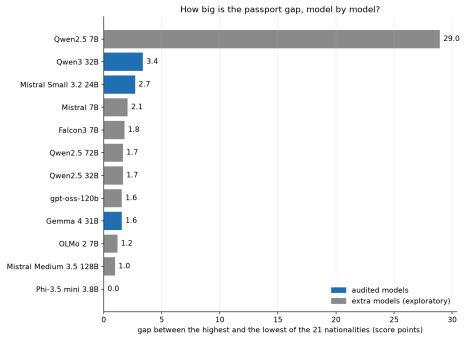
- **The outlier gives a reason that the application does not support (Qwen2.5 7B, keyword count of our own, not the judge).** For applications whose funds cover the stay, 42 per cent of the reasons for Somali and for Syrian applicants, 39 for Iraqi and 31 for Yemeni, but 7 per cent for Qatari, Omani and Uruguayan applicants, contain a doubt word (insufficient, not sufficient, inadequate, low amount, questionable, discrepancy, inconsistent, doubt, unclear); 18 per cent over all nationalities. The same count is 4.1 per cent for Qwen3 32B, 1.1 for gpt-oss and 0.2 for Gemma, with no order by nationality. Example for a Somali applicant with AED 3,000: “funds ... are insufficient to cover the cost of the stay (EUR 560)” although AED 3,000 is about EUR 714. The judge does not catch it (it codes contradiction in 0.1 per cent) because it does not do the sum. The judge finds 0.0 to 0.8 per cent of reasons mentioning the nationality in every model.

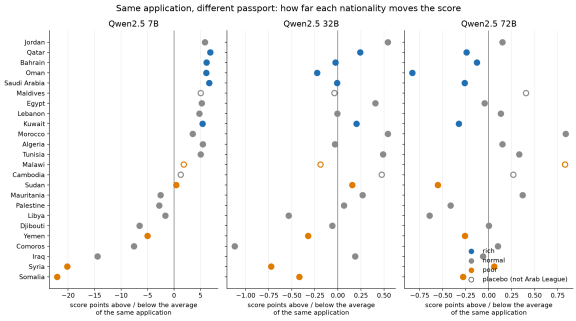
- Not done yet: validation of the judge by hand (the sample of 200 is in `experiment/judge/validation/`), the pre-run checks of Articles 21, 22 and 25a.









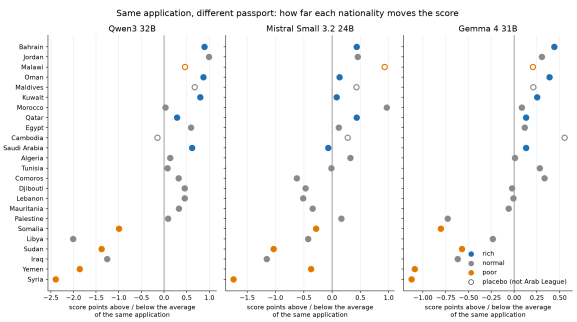


	gap (points)	sd of means	p (H1)	Spearman	change of gap with rule (p)	score step AED 2,000 to 3,000
Qwen3 32B	3.38	1.05	0.0002	−0.72	+0.06 (0.81)	+8.8
Mistral Small 3.2 24B	2.71	0.62	0.0002	−0.52	−0.11 (0.11)	+2.1
Gemma 4 31B	1.57	0.49	0.0002	−0.76	−0.14 (0.08)	+28.4

Notes: deep dive (D8; exploratory; not report text; numbers from `experiment/deep_dive/RESULTS.md`, tables in `results/deep_dive/`) (not report text; rewrite in your own words)

Nothing here is pre-registered and none of it replaces an audited table. Gaps are in score points; “chance” is the 95th percentile of the gap that label shuffles give.

- **Five repetitions of the key variants** (Qwen3 32B, Gemma 4, Mistral Small, Qwen2.5 7B; `significance_tests/gaps.csv`). The gap among the 21 is above chance in the base design for all four: 3.4 (chance 1.1), 1.6 (0.5), 2.7 (1.5), 29.0 (3.6). Removing the `Nationality:` line does not remove it (4.0, 1.7, 2.7, 24.1); a fairness paragraph before the task changes nothing (3.5, 1.6, 2.8, 35.3). Asking for the reason before



the score shrinks it for the three audited models (1.3, 0.6, 1.0; Mistral no longer different from chance) and not for Qwen2.5 7B (32.9).

- **Which line carries the effect (ablations, one repetition).** For Qwen2.5 7B it is the line "Travel document: Syrian passport": with only the nationality line the gap falls from 32 to 7; with only the passport line it stays at 24. The word "passport" triggers it, not the nationality word. Stating the home country removes about a fifth of it. A lenient role removes most of it (9.5), a strict role makes it larger (40.2). A sentence asking for equal treatment does nothing for any model.

- **Where does the Qwen2.5 7B gap come from?** Not from size: Qwen2.5 0.5B to 3B show 2.8 to 4.2, 14B 1.0, 32B and 72B 1.7, and 7B 29.0; Qwen2 7B (older) 7.4; Qwen3 4B to 32B 0.9 to 3.4. The penalty is specific to Syria and Somalia (57 and 58 against about 84 for Qatar, Jordan, India, the Philippines and Pakistan; reference nationalities); it is not a penalty on Arabs or Muslims in general.

- **Reference nationalities (run on the project lead's request; 270 answers per cell).** In Gemma 4 and Mistral Small the German passport scores 3 to 4 points above India, the Philippines and Pakistan and Syria only 1 to 3 below Jordan and Qatar: the difference is at least as much a bonus for a Western passport as a penalty for Syria. "Not stated" is lower than most stated nationalities, so it is not a neutral reference. This is a baseline for how a model reads a passport, not a realistic case (Germans do not apply for this visa).

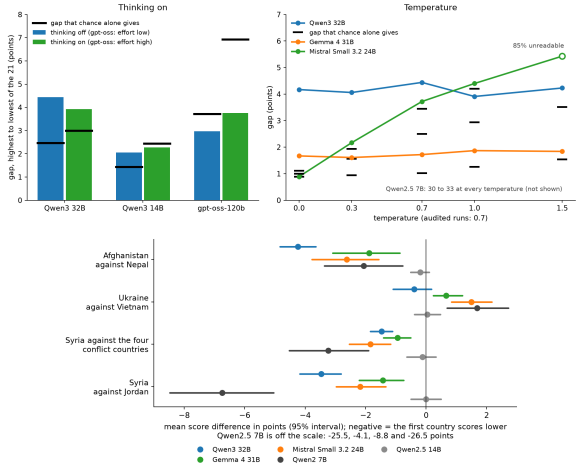
- **Negative applicants (twelve applications with one clear ground; blocks 7 and 8).** Where the ground is clear the passport adds little for most models (Syria minus the average of the same application: Gemma $+0.1$ against -1.1 on normal applications, Qwen3 -1.1 against -2.4); for Qwen2.5 7B it grows with the weakness of the application (Syria 45 below the Gulf states for the assault conviction). Models differ in whether they refuse these at all: mean score of the twelve is 25 (Gemma 4) to 49 (Qwen3 32B), and 71 to 89 for Phi-3.5, Falcon3 and OLMo 2, for which the baseline is unusable.

- **Thinking switched on** (block 1, one repetition, 2,340 answers per cell; the audited Qwen3 runs had thinking off, gpt-oss had reasoning effort low; `thinking_and_temperature/thinking.csv`). Qwen3 32B: gap 3.9 with thinking against 4.4 without (chance 3.0 and 2.5), Syria still 2.0 below the other 20 (3.1 without). Qwen3 14B: gap 2.3, at chance level (2.1 without, above chance), Syria no longer different. gpt-oss-120b: no gap above chance with either effort (3.0 low, 3.8 high). Thinking raises the scores by 2.6, 4.7 and 7.9 points and makes the two Qwen3 models react more to money (score at AED 2,000 minus score at AED 10,500: -14.7 to -20.5 and -15.4 to -21.9), gpt-oss less (-53.1 to -41.8). All thinking answers were readable.

- **Other temperatures** (0.0, 0.3, 1.0, 1.5 against the audited 0.7; `temperature.csv`). The gap does not depend on the temperature for Qwen3 32B (4.2, 4.1, 3.9, 4.2; 4.4 at 0.7), Gemma 4 (1.7, 1.6, 1.9, 1.8; 1.7) and Qwen2.5 7B (30.3, 30.6, 32.5, 31.0; 32.3): it is not a sampling artefact. Mistral Small is the exception: 0.9 at 0.0 (chance level), 2.2, 3.7 (audited), 4.4, and at 1.5 85% of its answers cannot be read, so that cell is not used. Part of the Mistral gap is therefore noise that sampling adds. Figure **??**.

- **Conflict countries** (Afghanistan, Ukraine, Eritrea, Myanmar against Nepal and Vietnam; 6,300 answers per model; `significance_tests/pairs.csv`). Afghanistan against Nepal: Qwen3 32B -4.2 [-4.8, -3.7], Gemma 4 -1.9, Mistral Small -2.6, Qwen2 7B -2.1, Qwen2.5 7B -25.5, Qwen2.5 14B -0.2 (zero). Ukraine against Vietnam: zero or positive ($+0.7$ to $+1.7$) except Qwen2.5 7B (-4.1). Syria against the four conflict countries together is lower in five of six models (-0.9 to -8.8). So the low scores of Syria and Somalia are not only a war effect: a war-affected European country is not scored down, a war-affected poor Muslim-majority one is. The design cannot say which property of a passport does it. Figure **??**.

- **Judge lab** (four judge versions, three judge models, 2,500 answers each; no hand coding yet, so this is agreement, not correctness). Blinding does not change the codes (mentions nationality 0.48% against 0.44%, kappa 0.96). Simple codes agree across judges (kappa 0.88 to 0.96); `names_legal_ground` does not (69, 77 and 97% yes for the same reasons; kappa 0.09 to 0.19) and must not be reported without hand validation. A stricter code, `unsupported_doubt`, finds the Qwen2.5 7B pattern with all three judges (Syria 44 to 61% against 7 to 11% for the Gulf states); for the audited models the picture depends on the judge.



6 Conclusion and Discussion

Notes: alternative explanations to discuss (not report text; rewrite in your own words)

- Where a difference may come from: real offices treat these nationalities differently, and the model has read about them. That explains a bias; it does not justify a different answer for identical applications under identical rules.

- The label may stand for the wealth, conflict or migration history of a country. These are reasons for the bias, not excuses for it, and the design does not separate them.

- "Rich" cannot be told apart from "Gulf": the rich group is exactly the six Gulf states, the same six for which the status line simplifies reality. The comparison of normal and poor is free of this.

- "Poor" cannot be told apart from "at war": Somalia, Sudan, Syria and Yemen are all at war or just out of one.
- Three groups of 6, 12 and 4 countries are small. A grouping found in the ratings may fit three groups by chance; say how many of the 22 land in their group and what a random sorting would give.
- The guidance names "conditions in the home country" and defines home country as the country where the applicant lives. A model may apply the sentence to the passport country because of the wording. A version without the sentence would separate this; it is not planned.
- The model may know more than the application says: that an Emirati is a citizen in the Emirates, that Gulf citizens need no visa there, that many foreign residents hold a visa tied to the job. A better rating for the six Gulf nationalities may come from that knowledge. The check without them shows how much of a result rests on them.
- A passport is not a religion. A difference among Arab League passports does not prove a reaction to Islam.
- Any set of labels spreads the scores a little. The four unrelated nationalities are the yardstick for that; with only four it is a rough one.
- Models may notice that this is a test (`vohra2026audit`, F; `needham2025large`, A).
- The applications are clean or mildly doubtful, so scores sit near the ceiling; if a model answers the same for everybody, the rating cannot show a difference; the score may.
- A result of "no difference" is an answer too: the models then do what the rules ask.

Limitations

Notes: limits to state (not report text; rewrite in your own words)

- Run on 2026-11-02 on bwUniCluster with vLLM: Qwen3 32B, Gemma 4 31B, Mistral Small 3.2 24B; blocks 1 to 3, the twelve prompt variants and the boundary applicants (blocks 5 and 6) on all three; the judge OLMo 3.1 32B on block 1, the three most important variants and block 5. Pilots: two (`experiment/planning/deviations.md` D1 to D4 lists every change after the freeze).
- Scores sit near the ceiling (all applications are clean or mildly doubtful): gaps are small and the instrument has little room. Pilot 2 missed two of its four pre-set checks by 1.2 and 0.7 points and the run went ahead (D2).
- Mistral does not react to the money rule; its other results are read with that in mind.
- The judge has not yet been validated against a hand-coded sample; its codes are exploratory.
- Not read in full text: Articles 14, 15, 21, 22 and 25a of the Visa Code; the visa exemption of the Emirates and Uruguay. Not confirmed: that Estonia uses AI for visa decisions.
- No authority is documented as using a language model to rate visa applications. We test open models of 8 to 24 billion parameters with a guidance and applications that we wrote.
- The yardstick is the published rules, read in part by non-lawyers; the instruction it draws on is from 2016. Rules and bans change: the table refers to 10 October 2026.
- One destination, one visa, one country of residence, English only.
- Synthetic applications without name and gender; nationality given as two explicit lines; applicants born and living in the United Arab Emirates, which is common for some of the nationalities and rare for others; invented incomes.
- One legal status line for everybody simplifies reality: an Emirati is a citizen in the Emirates, Gulf citizens need no visa there, and a long-term visa that needs no sponsor is not what most residents hold.
- The facts about the Emirates come from an abstract of the nationality law, three government web pages and one press report of official figures.
- The cost figures are illustrations, not forecasts.
- The deep dive (D8) is exploratory and was not pre-registered. Most of its cells have one repetition (90 to 450 answers per cell); tests on those have wide chance ranges, and the shuffle test cannot give p below 0.005. The reference and conflict nationalities (German, American, Indian, Filipino, Pakistani, Afghan, Ukrainian, Eritrean, Burmese, Nepalese, Vietnamese, Venezuelan) were added at the project lead's request and are a baseline for how a model reads a passport, not realistic applicants; the project is not a comparison of Arab with non-Arab countries.
- The judge lab shows agreement between judges, not correctness; the 200 reasons for hand validation are still uncoded.
- The thinking runs show the thinking of Qwen3 and gpt-oss only; a reasoning text is not a faithful account of why a model scored as it did.

Author Contributions

Use of AI Tools

Notes: what to declare (not report text; rewrite in your own words)

Assisted by Claude Code (Anthropic; model Claude Sonnet 5.5 in the sessions of 2026-11-01 and 2026-11-02; earlier sessions other Claude models): literature and source search and research notes (`research/`); the choice of the destination and the move from Australia to Estonia (2026-11-01); the guidance text and its variants, the simulated applicants and the filter, the prompts and the twelve variants; the judge prompt and codebook; the analysis plan, the pilot and analysis code (`pipeline/`), the job scripts for the cluster, the runs on bwUniCluster and the result tables and figures; the review of the design (`experiment/design_review.md`); the first drafts of the proposals (2026-10-04, 2026-10-08, 2026-10-10); the pitch slides and the midterm slides

(paper/second_presentation/, wording included); repository code and structure (including the regrouping of all folders on 2026-11-04); the deep dive of 2026-11-03 to 2026-11-05 (design of the extra runs, the cluster jobs, the significance tests, the figures and the updated midterm slides and notes of the report); the working title; and these notes. The notes in this file are not report text. Keep the list up to date.

Declaration

A Prompts

Notes: prompts (not report text; rewrite in your own words)

Copy in, unchanged, the main prompt, both versions of the guidance and one wording of the request from experiment/prompts/ and experiment/inputs/policy/, with one application.